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Project Title:	A Novel Biodosimetry Method For Monitoring Radiation-Induced Genetic Damage		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
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Zip Code:	53711-5399	**Congressional District:**	2
Comments:			
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	
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No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA HQ
Contact Monitor:		**Contact Phone:**	
Contact Email:			
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	For extended space missions, exposure to hazardous radiation from galactic cosmic radiation and solar particle events poses a serious threat to the health of the astronauts. Dosimeters measure radiation dose received but not its biological effects or individual differences in radiation sensitivity. Accurate methods for measuring the extent of an individuals genetic damage resulting from accumulated radiation exposure is critical for estimating personal health risks and need for countermeasures. The overall goal of this proposal is to investigate the efficacy of a novel biodosimetry method for monitoring mutational load resulting from radiation-induced mutations in repetitive DNA sequences. Repetitive sequences constitute nearly one third of the human genome and some have been shown to be extremely sensitive to radiation-induced mutations. Most repetitive sequences are in non-coding regions of the genome and, therefore, alterations in these loci are usually not deleterious to the survival of the host cell. Thus, mutations in non-coding repetitive sequences can accumulate, providing a stable molecular record of DNA damage from all past exposures.		

Task Description:	Progenitor stem cells are long lived and continually divide producing identical duplicate progeny cells. Mutations in repetitive DNA loci in progenitor stem cells will accumulate and through analysis of progeny blood or sperm cells can serve as biological sensors to monitor radiation-induced genetic damage. To test this hypothesis we will (a) screen a broad range of highly mutable repetitive DNA sequences to identify genetic loci most sensitive to high-LET radiation-induced mutations, (b) show that radiation-induced mutations in progenitor stem cells are cumulative and that mutational load can be quantitatively measured by sampling a population of single or small pools of progeny blood or sperm cells for changes in repetitive DNA sequences, and (c) investigate possible mechanisms causing radiation-induced genomic instability in repetitive DNA loci. Knowledge obtained from completion of this proposal will determine the efficacy of a novel quantitative method for monitoring radiation-induced mutational load that would be applicable to both mouse models and humans. Mutational load profiling, through analysis of changes in repetitive DNA sequences, is a simple, non-invasive and generalized approach for monitoring an individual's cumulative record of radiation-induced mutations that would be useful in predicting and minimizing health risks.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Mutational load profiling, through analysis of changes in repetitive DNA sequences, is a simple, non-invasive and generalized approach for monitoring an individual's cumulative record of mutations that would be useful in predicting and minimizing health risks for astronauts or any individual exposed to ionizing radiation or chemical mutagens. Preliminary data suggest that the markers identified in this study are also sensitive biomarkers for oxidative stress and may be useful in early detection of conditions that could lead to cancer and other degenerative diseases and may be useful in the helping to develop effective countermeasures.
Task Progress:	Microsatellite markers containing of 1 to 6bp tandemly repeated units were screened for sensitivity to radiation-induced mutations in human cells (normal fibroblast, normal lymphoblast, and mismatch repair deficient cancer cell lines), mouse cells (immortalized C57Bl/6J fibroblast) and mice (SupFG1+/-, Gpx4+/-, Gpx4 transgenic, Msh2-/-, Msh2+/-). Cell cultures and mice were exposed to 0.5 to 3 Gy iron ions at NSRL, Brookhaven National Laboratory. Cells were grown for 72 after irradiation before harvesting and mice were maintained for 10 weeks after irradiation to allow turnover of sperm cells. Microsatellite analysis was performed using small-pool PCR with DNA equivalent to 1-2 genomes to detect mutant alleles present at low frequencies. To test whether repeat markers were sensitive to oxidative stress, C57Bl/6J mice were treated with paraquat to induce oxidative stress and DNA from livers assayed for mutations. In addition, cell cultures of normal human fibroblasts were treated with 0.01mM to 1.0mM hydrogen peroxide for 1 hour, allowed to recover for 3 days then screened for mutations. We have identified three classes of microsatellite markers that are sensitive to radiation-induced mutations in mouse and human cells. These include mononucleotide repeats with polyA runs of over 40bp (Figure 1), tri-, tetra- and penta-nucleotide repeats on the Y chromosome (Figure 2), and some A-rich pentanucleotide repeats (Figure 3). The length of the polyA tract in mononucleotide repeats was positively correlated with sensitivity to radiation and ROS-induced mutations. Mononucleotide repeats with polyA runs of 24 to 37bp (mBat-24, 26, 30 and 37) were not altered in cells or mice exposed radiation or oxidative stress (Table 1). In contrast, markers with longer polyA tracts (mBat-59 to 67) exhibited high level of instability under similar conditions. Selected markers for each of these three classes of repeats exhibited a dose dependent response to radiation and oxidative stress. The dose-response curve for mutation induction in normal human fibroblasts exposed to 0.5 to 3 Gy iron ions using 13 Y chromosome repeats was concave upwards at 3 Gy probalby due to saturation of the DNA repair system. Some repeats are clearly more sensitive to radiation and ROS-induced mutations than others. This increased sensitivity may reflect differences in secondary structure between repeat sequences and/or differences in chromatin structure that could affect accessibility of the ROS molecules to the DNA. Repeats on the Y chromosome exhibited greater sensitivity to radiation and ROS-induced mutations compared to similar repeats on the autosomal chromosomes. This increased mutability of Y chromosome repeats may reflect the lack of homologous recombination in the region of the Y chromosome where these markers are located. Mismatch repair does not appear to play a major role in repair of the radiation or ROS-induced mutations, as markers known to be very unstable in DNA mismatch repair deficient cells were completely stable following exposed to either mutagen. Our preliminary results indicate that mutational load profiling, through analysis of changes in repetitive DNA sequences, may be useful for monitoring an individual’s exposure to radiation and potential health risks. Additional experiments are underway to better understand how radiation induces mutations in some repetitive sequences and to identify an optimal set of markers for development of an effective multiplex assay for biodosimetry.
Bibliography Type:	Description: (Last Updated: 04/16/2019)
Articles in Peer-reviewed Journals	Bacher JW, Flanagan LA, Smalley RL, Nassif NA, Burgart LJ, Halberg RB, Megid WM, Thibodeau SN. "Development of a fluorescent multiplex assay for detection of MSI-High tumors." Disease Markers 2004;20(4-5):237-50. PMID: 15528789 , Nov-2004
Articles in Peer-reviewed Journals	Bacher JW, Abdel Megid WM, Kent-First MG, Halberg RB. "Use of mononucleotide repeat markers for detection of microsatellite instability in mouse tumors." Molecular Cancinogenesis (submitted), January 2005. , Jan-2005
Patents	Application #60/621277. Provisional patent application, October 2004. Oct-2004 Bacher J, Halberg R, Kent M. "METHODS AND KITS FOR DETECTING MUTATIONS."
Presentation	Bacher, J.; Halberg, R.; Megid, W.; Kent, M.; Prolla, T. "Monitoring Radiation-Induced Genetic Damage" 3rd International Workshop on Space Radiation Workshop and 15th Annual NASA Space Radiation Health Investigators’ Workshop. May-2004
Presentation	Bacher, J., Halberg, R., Megid, W., Kent, M., Prolla, T. "A Novel Biodosimetry Method for Monitoring Radiation-Induced Genetic Damage" 16th Annual NASA Space Radiation Health Investigators Workshop. May-2005
Presentation	Bacher, J.; Halberg, R.; Megid, W.; Prolla, T "A Novel Biodosimetry Method for Monitoring Radiation-Induced Genetic Damage" NASA Bioastronautics Investigators Workshop, Galvenston, TX Jan-2005